

Accelerating Electric Vehicle Adoption

A County-Level Framework for Targeted Infrastructure Investment





Targeted Allocation of Public dollars for EV Infrastructure & Policies

~28%

of 2022 U.S. greenhouse gas emissions came from transportation
(EPA — see Sources, slide 14)

This project builds a scalable, data-driven framework

- *using Washington State as a case study*
- *to identify optimal locations for EV related investments to maximize adoption & ROI*

OUR APPROACH

Two models, one goal: know where — and why — to invest



1. Unify Data

Five Washington State sources
(2017 – 2025) County/Year Level

FOUNDATION



2. Model Drivers

Ridge regression for
quantification

PREDICTION



3. Segment Counties

K-Means clustering for
segmentation

SEGMENTATION

Five state datasets, two modeling approaches



Datasets (WA State, 2017–2025)

Source	Provides
Vehicle Registrations	Registered EVs (numerator) & Total registered vehicles (denominator) by county
EV Population Data	EVs by Model Year & Attributes
Median Income (OFM)	County income
Population (OFM)	County population
Charging Stations (WSDOT)	Public charger locations

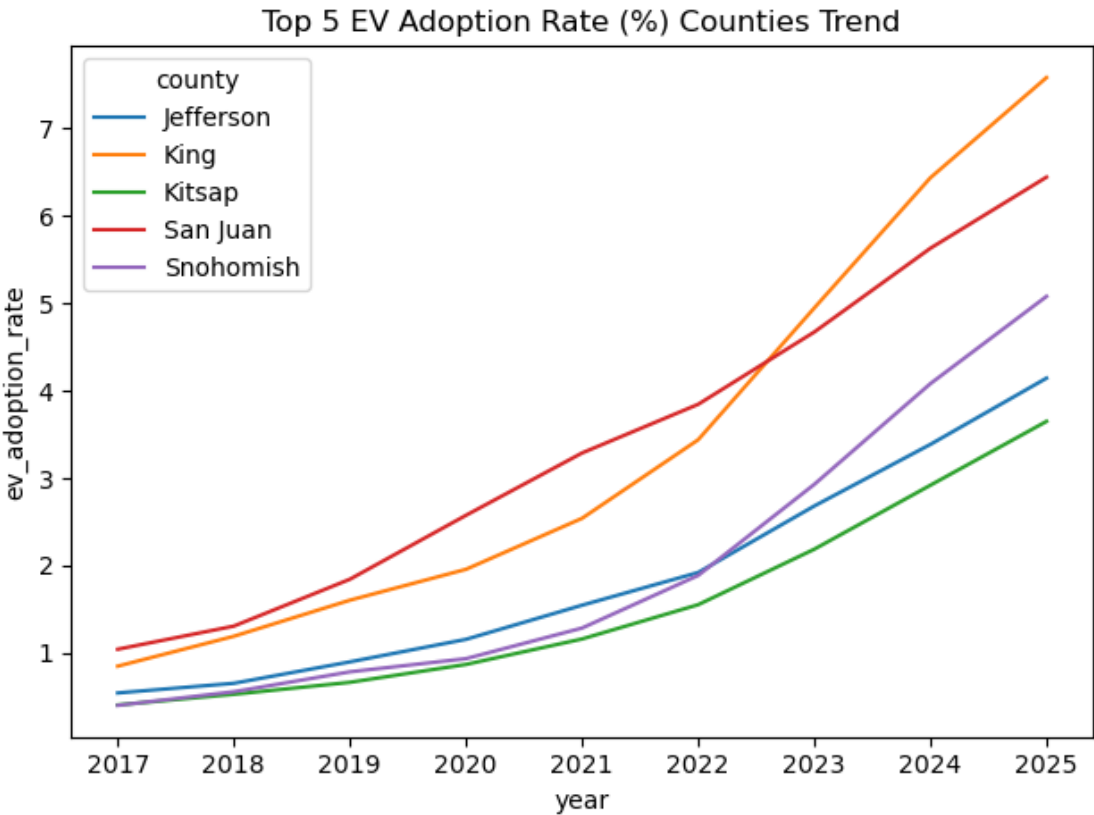


Modeling Approach

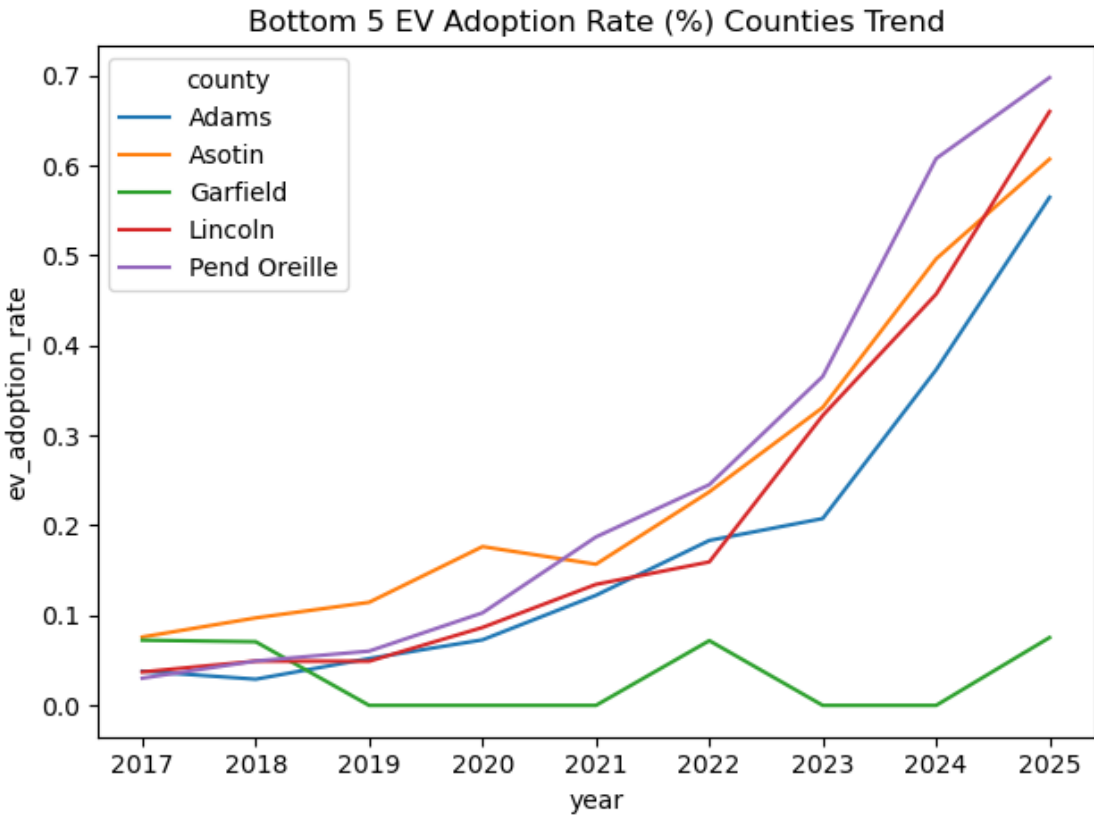
Model	Predicts	Key Inputs
Ridge Regression	County adoption rate	Prior adoption, income, charging density, population
K-Means Clustering	County segments	Income, charging density, population

WHAT THE DATA SHOWS

The adoption gap between counties is very wide



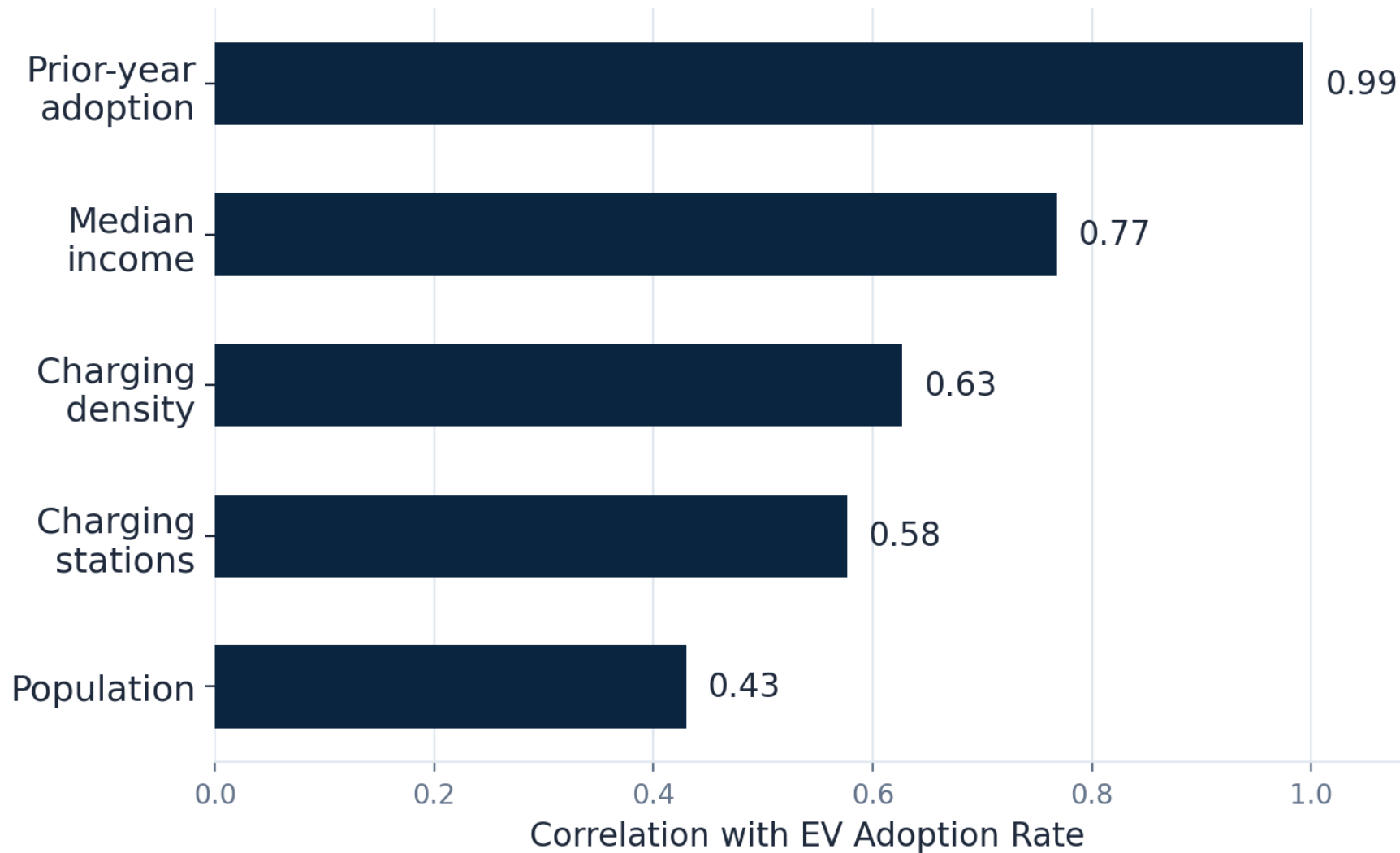
Top 5 counties by EV adoption rate, 2017–2025



Bottom 5 counties by EV adoption rate, 2017–2025

WHAT THE DATA SHOWS

Strong Correlations with Prior year adoption, Income, Charging Density



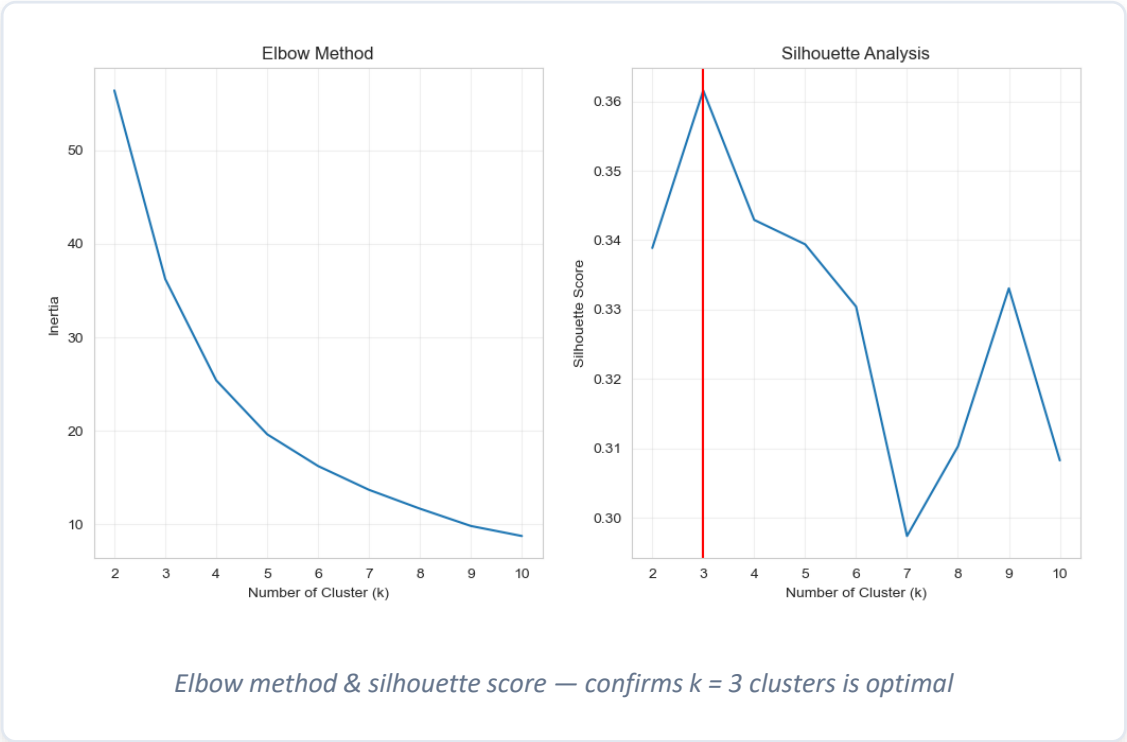
Both models were fitted & tested



Ridge Regression — Held-Out Test (2024–2025)

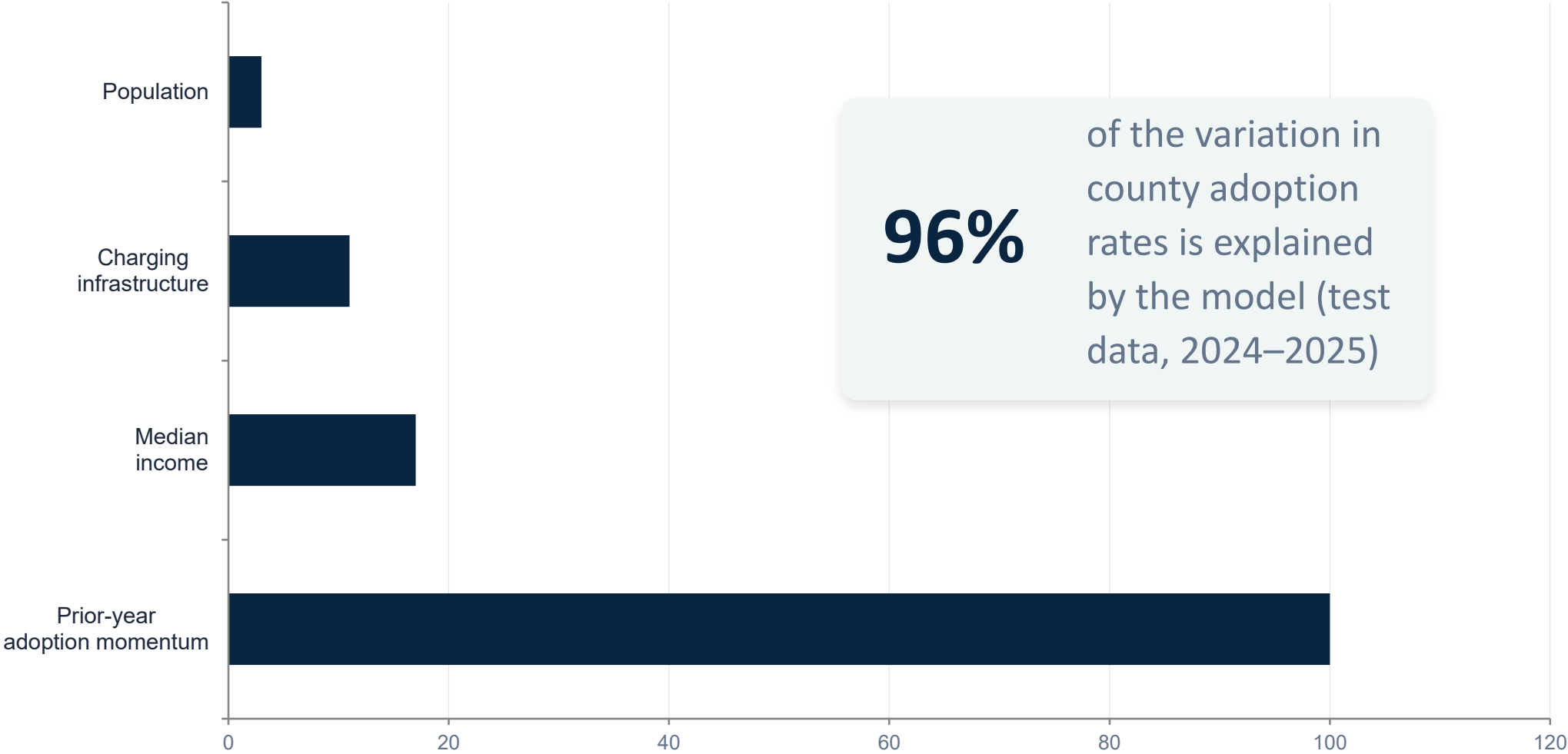
Metric	Value	What it means
R ²	0.96	Explains 96% of adoption variance
RMSE	0.14	Low prediction error
MAE	0.11	Predictions close to actual values

Validated with time-aware cross-validation (5-fold TimeSeriesSplit) — the model was never shown future years during tuning.



KEY FINDING 1

Momentum, income & infrastructure — Key Drivers



KEY FINDING 2

Washington's counties fall into three distinct adoption profiles



High-Adoption Leaders

16 counties · 41%

~3.0%

avg. adoption rate

Income ~\$96K

Charging density Low

King, Snohomish, Clark, and other high-income, high-population counties. Adoption is self-sustaining.



Rising Markets

22 counties · 56%

~1.6%

avg. adoption rate

Income \$60K–95K

Charging density High

Columbia, Lincoln, Adams, and similar counties. Infrastructure is compensating for lower income.



Charging Desert

1 county (Garfield)

~0.1%

avg. adoption rate

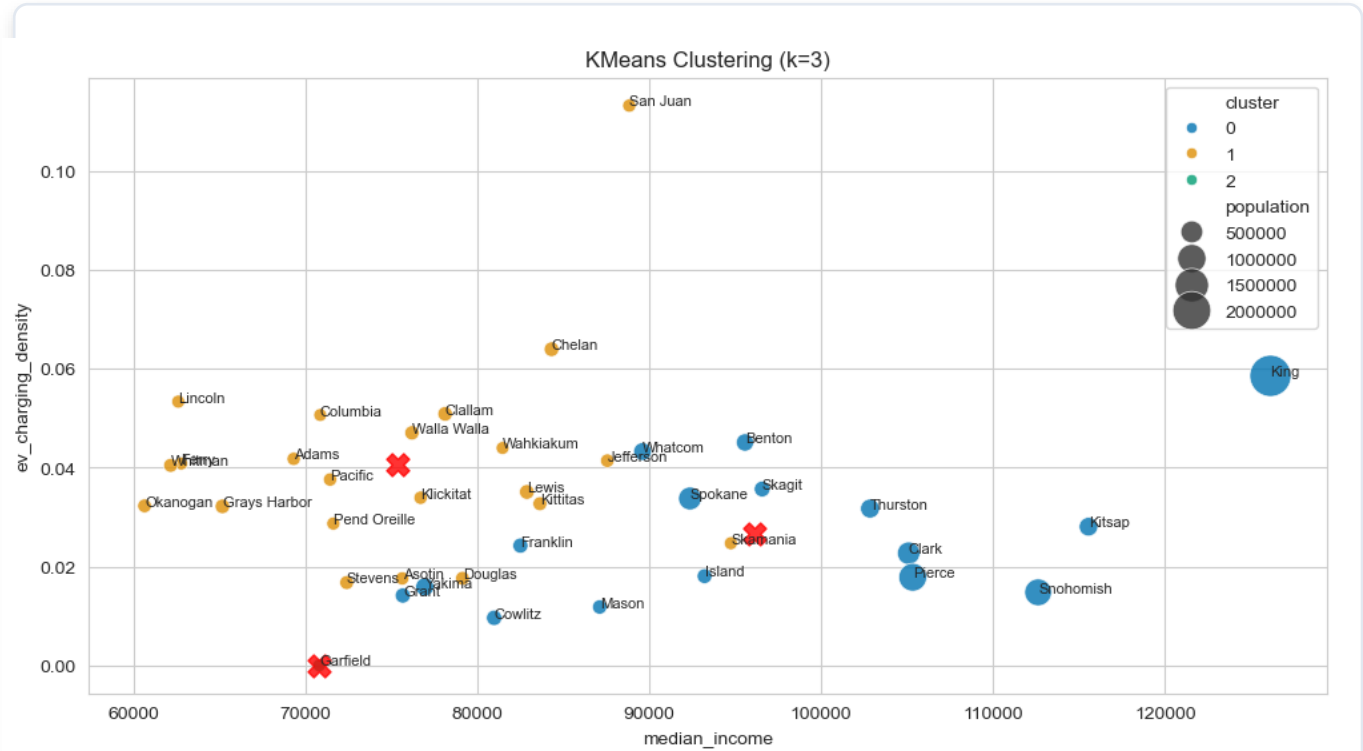
Income ~\$71K

Charging density Zero

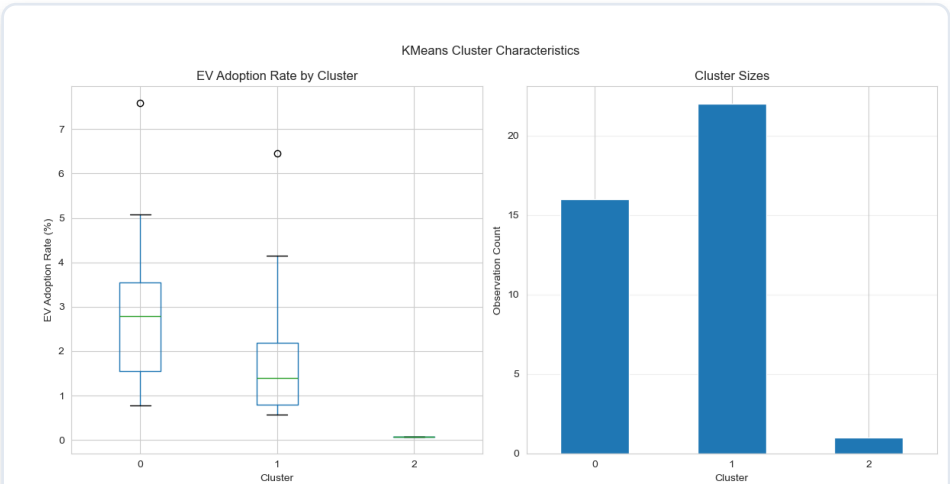
Comparable income to Rising Markets counties, but zero public charging — momentum never starts.

SEGMENTATION VISUALIZED

Income and infrastructure separate the three groups effectively



Counties plotted by income vs. charging density, sized by population;
× marks cluster centers



Adoption rate and county count by cluster —
confirms the 3 groups are meaningfully different

THE RECOMMENDATION

A three-phase roadmap, matched to funding source

1

PHASE 1
Foundation

GARFIELD COUNTY

Seed baseline highway-corridor charging to eliminate the state's only total charging desert.

FUNDING

100% public capital

2

PHASE 2
Activation

RISING MARKETS COUNTIES

Vehicle purchase rebates and used-BEV subsidies to unlock pent-up demand; maintain fast-charger uptime.

FUNDING

Public rebates + private matching

3

PHASE 3
Autonomy

HIGH-ADOPTION LEADERS

Rely on private commercial capital for destination charging; public spend limited to grid capacity and select low-income pockets.

FUNDING

Mostly private capital

WHY IT MATTERS

Concentrate scarce public capital where it changes outcomes



Closes the state's one true charging desert

A single targeted investment in Garfield County removes the only structural barrier where infrastructure — not income — is the constraint.



Activates demand where infrastructure is ready

Rising Markets counties already have charging supply; rebates convert existing infrastructure into adoption.



Frees public capital for where it's needed most

High-Adoption Leaders can be served by private commercial investment, redirecting public dollars elsewhere.



Built to scale beyond Washington

The same data pipeline and models can be re-run on any state with comparable registration and infrastructure data.

CONCLUSION

The data points to a clear, fundable strategy



EV adoption is not random — it is driven, predictably, by momentum, income, and infrastructure. Knowing which of those three is binding in a given county tells us exactly what kind of investment will move the needle there.

Regression

Quantifies what drives adoption and by how much ($R^2 = 0.96$).

Clustering

Turns those drivers into 3 actionable county segments.

Roadmap

Matches each segment to the right funding mechanism.



NEXT STEPS

Recommended path forward

- 1 Pilot a highway-corridor charging investment in Garfield County
- 2 Design a rebate / used-BEV subsidy program / (infrastructure as needed) for Rising Market counties
- 3 Engage private and utility partners on buildout in Leader counties
- 4 Extend the framework to additional states for broader validation

Data sources and references



Datasets (Washington State Open Data, 2017–2025)

Dataset	Source
Electric Vehicle Population Data	data.wa.gov/Transportation/Electric-Vehicle-Population-Data/f6w7-q2d2
Vehicle Registrations by Class and County	data.wa.gov/Transportation/Vehicle-Registrations-by-Class-and-County/hmzg-s6q4
Median Household Income Estimates	ofm.wa.gov/data-research/economy/median-household-income-estimates
Population Estimates	ofm.wa.gov/data-research/population-demographics/estimates
EV Charging Stations Data	ev-map-wsdot.hub.arcgis.com

Additional references

U.S. EPA, "Sources of Greenhouse Gas Emissions" — transportation share of 2022 U.S. emissions (epa.gov/ghgemissions/sources-greenhouse-gas-emissions), cited on slide 3.

American Lung Association Video on Electric Vehicle Solution — <https://www.youtube.com/watch?v=V5w2TiEohs8> on slide 2.

Full methodology, assumptions, and limitations are documented in the accompanying project report and README.



Thank You

Questions & discussion



gawandesaillee@gmail.com



[GitHub Repository Link](#)



[linkedin.com/in/saileegawande](#)